

EXPERIMENT 7A

DATA VISUALIZATION USING TABLEAU – Sample Dataset

Aim:

To connect, analyze, visualize, and create interactive dashboards using Tableau for effective business intelligence and data analytics.

Sample Dataset:

Student Performance Dataset

Student	Department	Marks
Arun	CSE	85
Bala	IT	78
Divya	CSE	95
Priya	AIDS	90
Ravi	IT	72
Karthik	AIDS	88

Algorithm 1:

Connecting Data Source

Step 1 Open Tableau Desktop

Step 2 Select Connect

Step 3 Choose Excel File

Step 4 Browse Dataset

Step 5 Open Dataset

Step 6 Verify Imported Data

Algorithm 2:

Creating Worksheet

Step 1 Open Sheet 1

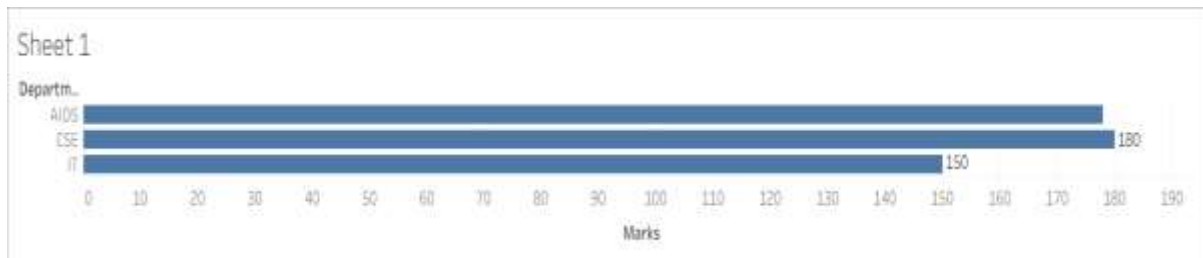
Step 2 Drag Department to Columns

Step 3 Drag Marks to Rows

Step 4 Select Visualization

Step 5 Analyze Results

Output:



Algorithm 3:

Bar Chart Creation

Step 1 Drag Student → Columns

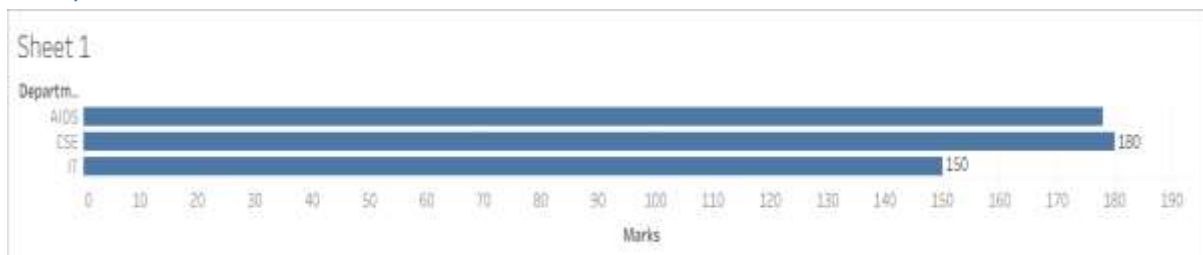
Step 2 Drag Marks → Rows

Step 3 Select Bar Chart

Step 4 Apply Labels

Step 5 Format Visualization

Output:



Algorithm 4:

Pie Chart Creation

Step 1 Drag Department

Step 2 Drag Marks

Step 3 Select Pie Chart

Step 4 Show Labels

Step 5 Analyze Distribution

Output:

Department-wise Marks Contribution

Contribution of departments to total marks.



Algorithm 5:

Line Chart Creation

Step 1 Drag Student

Step 2 Drag Marks

Step 3 Select Line Chart

Step 4 Apply Formatting

Output:

Student Performance Trend

Marks trend among students.



Algorithm 6:

Creating Calculated Field

Objective

Calculate Percentage

Marks Formula (Marks / 100) * 100

Procedure

Analysis → Create Calculated Field

Name: Percentage

Formula: [Marks]

Output:

Departm..	Student	Marks	Percenta..
AIDS	Karthik	88.00	88.00%
	Priya	90.00	90.00%
CSE	Arun	85.00	85.00%
	Divya	95.00	95.00%
IT	Bala	78.00	78.00%
	Ravi	72.00	72.00%

Algorithm 7:

Applying Filters

Step 1 Drag Department to Filters

Step 2 Select Department

Step 3 Apply Filter

Step 4 Observe Updated Charts

Algorithm 8:

Creating Parameter

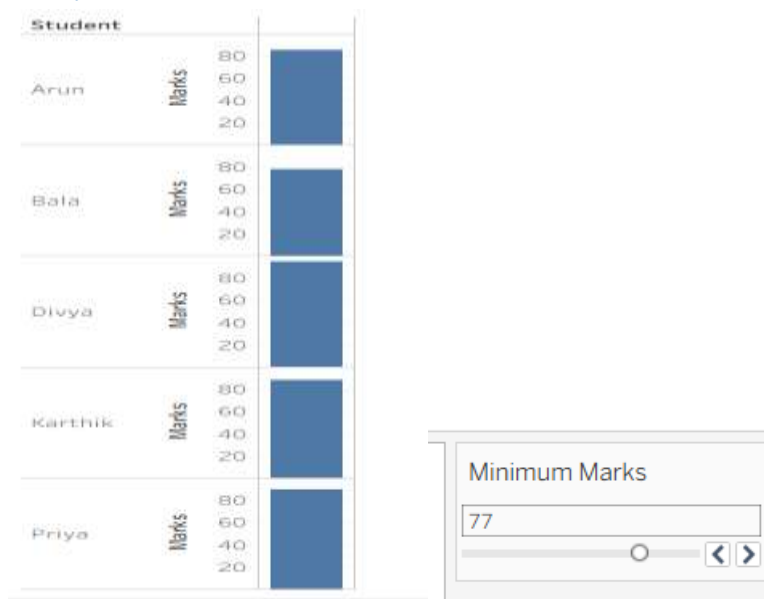
Step 1 Create Parameter

Step 2 Specify Minimum Marks

Step 3 Apply Parameter

Step 4 Display Matching Records

Output:



Algorithm 9:

Dashboard Creation

Step 1 Create Multiple Worksheets

Step 2 Select Dashboard

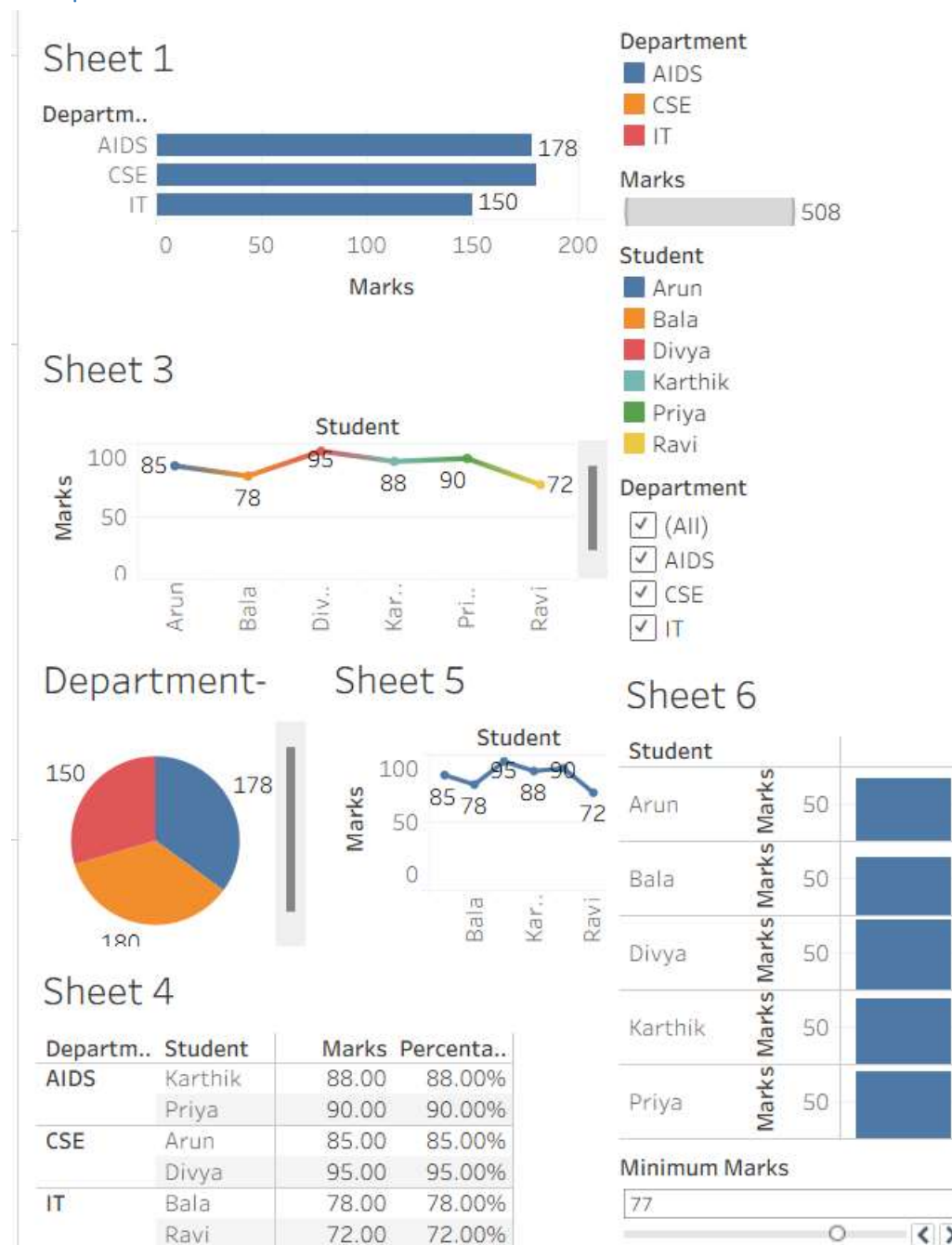
Step 3 Drag Worksheets

Step 4 Arrange Components

Step 5 Add Filters

Step 6 Save Dashboard

Output:



Result:

Thus, Tableau was used successfully to connect datasets, create worksheets, apply filters and parameters, generate visualizations, build dashboards, and create stories for business intelligence and analytical decision-making.